MAT 221 - CALCULUS I
Lesson 20: - L'Hopital's Rule and Indeterminate Forms

DR. M. KOORAPETSE and DR. T. YANE

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University of Botswana
Department of Mathematics

INDETERMINATE FORM

We recall that if limits $\lim_{x\to x_0} f(x)$ and $\lim_{x\to x_0} g(x)$ both exist, then

$$\lim_{x\to x_0} \frac{f(x)}{g(x)} = \frac{\lim_{x\to x_0} f(x)}{\lim_{x\to x_0} g(x)}, \tag{1}$$

provided $\lim_{x\to x_0} g(x) \neq 0$. However, in computing these limits, we can encounter outputs of the form $\frac{0}{0}, \frac{\infty}{\infty}, 0^0, 0^\infty$ etc called indeterminate form. Examples follow in the table below.

LIMITS	INDETERMINATE FORM
$\lim_{x\to 0} \frac{\sin x}{x}$	$\neq \frac{0}{0}$
$\lim_{x\to\infty} \frac{3x^2-2x+4}{x^2+4x+4}$	$\neq \frac{\infty}{\infty}$
$\lim_{x\to 0^+} (\csc x - \cot x)$	$\neq \infty - \infty$
$\lim_{x\to 0^+} x \ln x$	$\neq 0.\infty$
$\lim_{x\to 0^+} x^{ax}$	0^0
$\lim_{x\to\infty} x^{\frac{1}{x}}$	∞^0
$\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^x$	$\neq 1^\infty$

In the case of indeterminate cases $\frac{0}{0}, \frac{\infty}{\infty}$, we apply L' Hopital's rule, which is stated by the theorem that follows.

Theorem 1

Suppose $f(x)$ and $g(x)$ are continous functions on $x \in \mathbf{R}$ and differentiable at x_0, if $\lim_{x \to x_0} \frac{f(x)}{g(x)}$ is indeterminate $\frac{0}{0}$ or $\frac{\infty}{\infty}$ satisfying

$$\lim_{x \to x_0} \frac{f^{(n)}(x)}{g^{(n)}(x)} = L \in \mathbf{R}, n \in \mathbf{N}$$

Example Evaluate the following limits:

(a) $\lim_{x \to 1} \dfrac{x^3 - 5x^2 + 4}{2 - x^3 - x^4}$

(b) $\lim_{x \to 0} \dfrac{\sin x}{x}$

(c) $\lim_{x \to 0} \dfrac{\cos x - 1}{x}$

(d) $\lim_{x \to 0} \dfrac{\sin x}{x^3}$

(e) $\lim_{x \to 0} \dfrac{\sin 3x - 3x}{x^3}$

Solutions

(a) $\lim_{x\to 1} \dfrac{x^3-5x^2+4}{2-x^3-x^4} \neq \dfrac{0}{0}$, *indeterminate form*

Apply L'Hopital's rule;

$$\lim_{x\to 1} \frac{3x^2-10x}{-3x^2-4x^3} = \frac{-7}{-7} = 1.$$

(b) $\lim_{x\to 0} \dfrac{\sin x}{x} \neq \dfrac{0}{0}$

Apply L'Hopital's rule;

$$\lim_{x\to 0} \frac{\cos x}{1} = \frac{1}{1} = 1.$$

(c) to (e) class exercise.

Example 2 Calculate the following limits

(a) $\lim_{x\to 1} \dfrac{\ln x}{x-1}$ (b) $\lim_{x\to\infty} \left(1+\dfrac{1}{x}\right)^x$

Solution

(a) $\lim_{x\to 1} \dfrac{\ln x}{x-1} \left(\neq \dfrac{0}{0}\right)$.

Apply L' Hopital's rule.

$\lim_{x\to 1} \frac{\frac{1}{x}}{1} = 1.$

Example 3 Calculate the following limits.

(a) $\lim_{x\to 0^+} x \ln x$ (b) $\lim_{x\to 0^+} x^{ax}$ (c) $\lim_{x\to 0} \dfrac{6x}{3x+2\sin x}$

Solutions

(a) $\lim_{x\to 0^+} x \ln x = \lim_{x\to 0^+} \dfrac{\ln x}{\frac{1}{x}}, \left(\neq \dfrac{-\infty}{\infty}\right)$

Apply L'Hopitals rule

$= \lim_{x\to 0^+} \dfrac{\frac{1}{x}}{\frac{-1}{x^2}} = \lim_{x\to 0^+} (-x) = 0.$

(b) $\lim_{x\to 0^+} x^{ax} (a \neq 0) \neq 0^0$, *Indeterminate form*

Let $y = x^{ax}$ that is, $\ln y = ax \ln x = \frac{a \ln x}{\frac{1}{x}}$

$$= \lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} \frac{a\ln x}{\frac{1}{x}}, \left(\neq \frac{-\infty}{\infty}\right)$$

Apply L' Hopital's rule

$$= \lim_{x\to 0^+} \frac{a\left(\frac{1}{x}\right)}{\frac{-1}{x^2}}$$

$$= \lim_{x\to 0^+} (-ax) = 0.$$

Since, $\lim_{x\to 0^+} \ln x^{ax} = 0$

therefore $= \lim_{x\to 0^+} x^{ax} = e^0 = 1$

Exercises Calculate the following limits:

(a) $\lim_{x\to 1} \dfrac{x^3 + x^2 - x - 1}{x^2 + 2x - 3}$

(b) $\lim_{x\to 0} \dfrac{\cosh x - e^x}{x}$

(c) $\lim_{x\to 0} \dfrac{x^2 - \sin 3x}{x^2 + 4x}$

(d) $\lim_{x\to 0} \dfrac{\tan x - x}{\sin x - x}$

(e) $\lim_{x\to 0^+} \dfrac{\sqrt{x}}{\sin 3\sqrt{x}}$